

From Paper to Pixel: Demonstrating the Power of Non-Digital Serious Games for Informing Digital Health Solutions

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Abstract

This study challenges the dominant use of digital formats when designing serious games to inform digital solutions, showcasing the unique advantages of using a non-digital game approach. While digital formats are often the default in HCI and serious games, our findings highlight how a paper-based game can capture diverse perspectives, foster engagement, and uncover nuanced insights often overlooked by digital-only approaches. Focusing on remote care for eating disorders, we co-developed a game with service users and providers to explore barriers to effective support and collaboratively design solutions. This inclusive, participatory process informed the creation of a digital toolkit tailored to stakeholders' needs. Our research broadens the scope of serious games within HCI and digital healthcare, demonstrating their utility beyond education, into service design and innovation. It underscores the value of applying non-digital games even when the target outcome is a digital intervention.

CCS Concepts

• **Human-centered computing** → Interaction design; Interaction design process and methods; Participatory design; Accessibility; Accessibility design and evaluation methods; • **Applied computing** → Life and medical sciences; Consumer health.

Keywords

Serious games, non-digital games, user-centered design, eating disorders, remote healthcare, HCI, co-design, digital health, accessibility, sensitive topics

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1 Introduction

Serious Games (SGs) are defined as games created for purposes beyond entertainment, such as training and education. They leverage gaming's engaging aspects to facilitate learning, skill development and/or behaviour change [5, 50]. This study demonstrates their broader utility by exploring how non-digital SGs can inform digital services.

Many researchers focus on digital SGs, limiting their scope to apps and immersive environments [35]. Systematic reviews have noted a void in the literature around non-digital games [26, 46, 51]. We argue that the choice of format should be driven by the specific context, goals and target audience. Cosimini et al. [10] also argue for a broader perspective, drawing attention to Abt's [2] original 1970s definition of SGs, which did not limit to digital formats. Our research details the co-development and application of a non-digital SG to inform digital solutions for remote Eating Disorder (ED) support, explaining how the non-digital format facilitated more inclusive digital service solutions.

EDs are sensitive topics, requiring research methodologies that minimise discomfort and promote engagement [28, 34, 36]. Creative, playful methodologies, such as non-digital SGs, can encourage broader participation and accessibility. We demonstrate this by evidencing the game's subsequent role in the co-creation of an open-access, digital healthcare toolkit. This research expands the application of non-digital SGs, highlighting their value in facilitating discussions, uncovering barriers, and co-developing solutions with diverse user groups. This is particularly relevant within the fields of HCI, SGs, and digital health where digital formats become the default. We encourage broader recognition that in some contexts, non-digital tools can provide more inclusive methods to inform digital solutions.

2 Related Work

2.1 Benefits & Applications of Serious Games

SGs can increase engagement, accessibility and information retention [21, 38]. They are particularly valuable when

developed to engage with topics from a third-person perspective, especially when exploring sensitive issues such as those that can be distressing, embarrassing, controversial or stigmatised [49]. Talking in the third person can help individuals to manage their emotions more effectively, whilst providing emotional distance to allow for more objective self-reflection [17, 25, 31]. Secondly, considering scenarios from a third-person perspective can help

individuals to recognise other's perspectives [18]. Thirdly, talking in the third person can take the focus away from the individual helping them to feel less anxious about participating and reducing their anxiety, which in turn can help them feel more comfortable to disclose information. This also provides control over their degree of self-disclosure as they are not required to reveal which contributions relate to first-hand experience [7]. Third person approaches therefore represent a more sensitive, less confronting approach to support individuals to talk about topics which they may otherwise avoid or feel uncomfortable discussing [25, 45].

SGs have been applied in many health contexts, e.g., depression [14, 41], anxiety [1], schizophrenia [48], affective disorders and phobias [3]. They have been shown to be effective in preventing, reducing and treating mental health disorder symptoms and increasing engagement with services [14, 26, 41]. They have also been used for training healthcare staff [29]. Whilst most SGs have an educational or behaviour change focus (e.g., adherence to treatment [8]), SGs can also be used as effective tools to engage service users with service design improvements [6, 16]. For example, in 2011 the Pennine Care NHS Foundation Trust [13] used a monopoly-style board game to engage low secure mental health service users in the refurbishment of their environment (e.g., room designs, colour schemes). The game was found to be an effective way to overcome barriers to meaningful service user involvement and led to tangible changes in their environment. Our SG also involved service users in co-creating solutions. However, we focused on experiences of remote care applicable across a range of contexts and services rather than within a singular physical healthcare space.

2.2 Digital vs. Non-Digital

Digital SGs are widely used and can offer benefits such as increased immersion, particularly with technologies like virtual reality [9, 11]. However, non-digital formats are often overlooked in HCI research [26, 46, 51]. This oversight is significant because non-digital SGs provide unique advantages, even when their purpose is to inform the development of digital interventions. Non-digital SGs are more accessible, fostering inclusion for those disadvantaged by digital barriers - a key stakeholder group when developing digital interventions [27, 43]. Additionally, non-digital formats, such as board games, offer tactile interaction, ideal for discussing sensitive topics in a safe environment. Face-to-face interaction enhances learning through shared play experiences, providing opportunities for players to reflect on behaviors and collaboratively solve problems [15]. Evidence suggests this process may have therapeutic benefits, such as combating isolation and alleviating stress through facilitating connections with others [30, 37]. A metaanalysis comparing digital and non-digital SGs found a higher effect size for non-digital games, suggesting they may be more efficient; with suggested contributing factors including interaction among peers, and greater flexibility of design and content [46]. These qualities make non-digital SGs a powerful tool for creating inclusive, user-centered interventions.

A non-digital format was important for accessibility in our work as this allowed us to engage a wider range of users. When designing remote healthcare interventions, it is imperative that individuals who could be the most disadvantaged from digital solutions are included in the process. Using a digital game in this context could

have inadvertently excluded individuals with limited digital access or literacy.

2.3 Eating Disorder Applications

Applications of SGs within EDs have been extremely limited, and those that do exist tend to be digital [47]. A 2022 review of 49 digital SGs in mental health for young people reported only one ED-specific game [12]. Outside of the SG field, the value of co-design and co-development is widely recognised for fostering inclusivity, with evidence showing that user-centered interventions are better suited to real-world challenges [4, 39, 40, 42–44]. Unfortunately, this consideration is often not extended to SG development, where involving users could significantly improve game relevance and appropriateness [20, 24]. Instead, many SGs are created using a designer-led approach which fails to involve users in the design of the game and/or assets [24]. A notable exception is the co-designed 'Maze Out' game [20], a digital SG piloted as a therapeutic tool for individuals undergoing ED treatment. Players navigated a metaphorical maze representing the complex, nonlinear journey of recovery. During which, they encountered challenges and decision points designed to reflect real-life obstacles in their treatment (e.g., coping with triggers, managing emotions, seeking support). The game was designed to help individuals practice their problem solving and coping strategies in a safe environment.

Our work shares some similar approaches with that adopted for the Maze Out game in that we involved people with lived experience (PWLE) of EDs and healthcare providers in the development of the game; helping to shape the game to ensure it resonates with users [20]. Our SG also follows a journey-style approach where players navigate a range of challenges designed to represent real world experiences. However, our work also differs in several important ways. First, our SG is not a therapeutic tool but a method to identify where interventions can enhance service experiences and/or provision (including aspects of digital literacy required to access remote care). Second, we opted for a non-digital format as including those who could be disadvantaged or most challenged by digital services was vital for our research (refer to 2.2). Third, we utilised a group game approach rather than a game designed to be primarily played individually. In doing so this facilitated group discussions (including between stakeholder groups), the pooling of expertise/experience, and the co-design of solutions. Fourth, the Maze Out researchers found that limited game content left some participants feeling unable to relate to some of the scenarios and/or feeling as if the game quickly became stagnant. We addressed this by involving players in leading the development of game character(s) and the challenges they encounter during the game. Lastly, we opted to develop our SG utilising a third-person perspective to facilitate sensitive discussions (refer to 2.1). Existing work highlights the need to be flexible when it comes to discussing EDs sensitively [32, 33], the game provided a less confronting method to facilitate discussions around digital barriers and potential solutions, without necessitating the sharing of personal information.

3 Method

The game was created as part of the Remote Healthcare for Eating Disorders throughout COVID-19 (RHED-C) project. Ethical

approval was granted by Northumbria University Psychology Department Ethics Committee [ref.45202].

3.1 Recruitment

Twenty participants were recruited as co-designers via the RHED-C participant pool, researchers' networks and advertising via social media. This included 17 PWLE and 3 service providers (1 National Health Service [NHS] ED service provider, 1 ED charity, and 1 mental health charity). All participants were UK residents, over 18yrs of age, with experience receiving and/or providing remote ED support. PWLE included those with an ED diagnosis and those who self-identified as having, or being in recovery from, an eating disorder. For safeguarding purposes, participants were excluded if they were currently receiving residential/inpatient care for EDs or had a post-discharge period of less than 6 months. The exclusion criteria were agreed based upon expert advice from an ED charity.

Fully informed consent was obtained prior to participation. Following participation, all individuals were fully debriefed and remunerated £50 for their time and contributions. Information for ED support services was provided at both the beginning and the end of the study (an important safeguard against participants not receiving this information if they were to drop out during the study).

3.2 Game Prototype

The initial game prototype was developed using a fictional character and storytelling approach, drawing inspiration from previous work demonstrating this as an effective approach for addressing sensitive topics [45]. This approach also allows individuals to self-distance (i.e., to take a step back when thinking about past experiences and reason from the perspective of a distanced observer); something which has been shown to facilitate the making of meaning from negative experiences [25]. In addition to reducing potential distress, the development of characters can facilitate deeper understanding of users' emotional and practical needs when it comes to service design or development [19, 23].

During the game, players work together to support a fictional service user through their remote healthcare journey, aiming to help them maximise the benefits of their remote care session(s). Throughout the game, the character encounters various barriers to accessing or receiving effective support. Players are tasked with completing the following:

- i) Character creation: At the beginning of the game, players create a character including information about their ED presentation (e.g., Robin, non-binary, white, age 23, distrusting of digital services) and the type of remote care they are accessing (e.g., body image therapy via Zoom).
- ii) Solutions to barriers: The team is randomly allocated a set of cards, each depicting barriers that may occur before, during and after the remote care session (e.g., instable internet connection, difficulty using platform functions). Players turn over one barrier card at a time and work together to identify potential solutions before progressing onto the next card.

The following sections provide more details about how the prototype was applied during the workshops to facilitate co-development of characters and the gameplay journey.

4 Findings

There were 2 phases of co-development, each phase consisting of 2 workshops (4 workshops in total) to test prototype feasibility, identify necessary revisions, and co-develop final game assets. In the first phase, we co-developed stock characters and expanded upon the barriers identified in the game. The second phase focused on co-developing the Journey Mat, a visual representation of how gameplay progresses.

Workshops were held in-person in North East England (2023-2024) and lasted 2 hours each. Workshops were audio recorded for transcription by the research team. To facilitate toolkit development, photographs were taken of co-designed work created during the sessions. To ensure participant confidentiality, privacy and comfort, no individuals were photographed or videoed.

The non-digital approach meant that all participants could take part equally, without digital literacy requirements or associated anxieties. Participants reflected positively on the non-digital approach and working collaboratively with others in-person. This experience, in itself, helped alleviate feelings of isolation that often affect individuals living with EDs:

"I found meeting up in person helpful because it helped me feel less isolated. Face-to-face meetings can be stressful at first but beneficial in that it gets me out of the house and interacting with the world a bit more." (P5)

The collaborative in-person approach had far-reaching benefits for those involved:

"The [in-person] workshop has given me an insight [due to] meeting other people who've had similar experiences [of remote care]. And that in itself has been therapeutic." (P2)

4.1 Phase 1: Co-developing Character Portraits and Barrier Cards

Whilst the game encourages players to create their own game character, we recognise that there are several instances when this may not be possible. For instance, players may find it difficult to initially get started, especially if it is their first time playing the game and/or they are feeling a little anxious at the start. There may also be instances where time is limited and therefore the process of character creating may reduce time available to capture the main game insights. For this reason, we sought to develop two stock characters which can be used as pre-made options when required. To achieve this, we asked participants (PWLE & healthcare professionals) in phase 1 of the workshops, to co-develop pen portraits for the characters using a custom-made template (Figure 1). The pen portraits included just enough information to act as a starting point for future players. Previous research has shown this provision of basic character details to be a successful approach for starting sensitive discussions without impeding participants' ability to add details based on their own experiences [45].

The in-person workshop provided a shared experience of creative, warm-up activities, whereby participants could chat over



Figure 1: Character pen portrait from one of the workshops

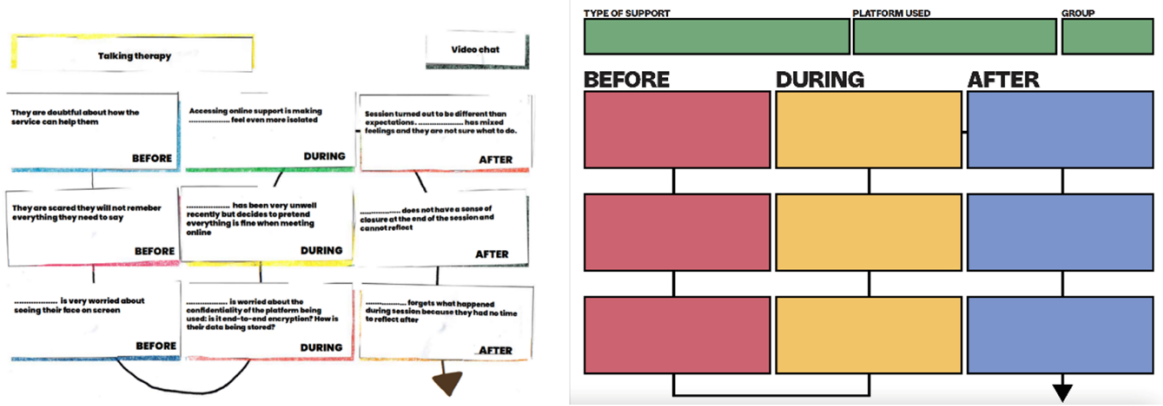


Figure 2: The game journey mat created during the workshops shown populated with some of the digital barrier cards (left) and the final version without barrier cards in place (right)

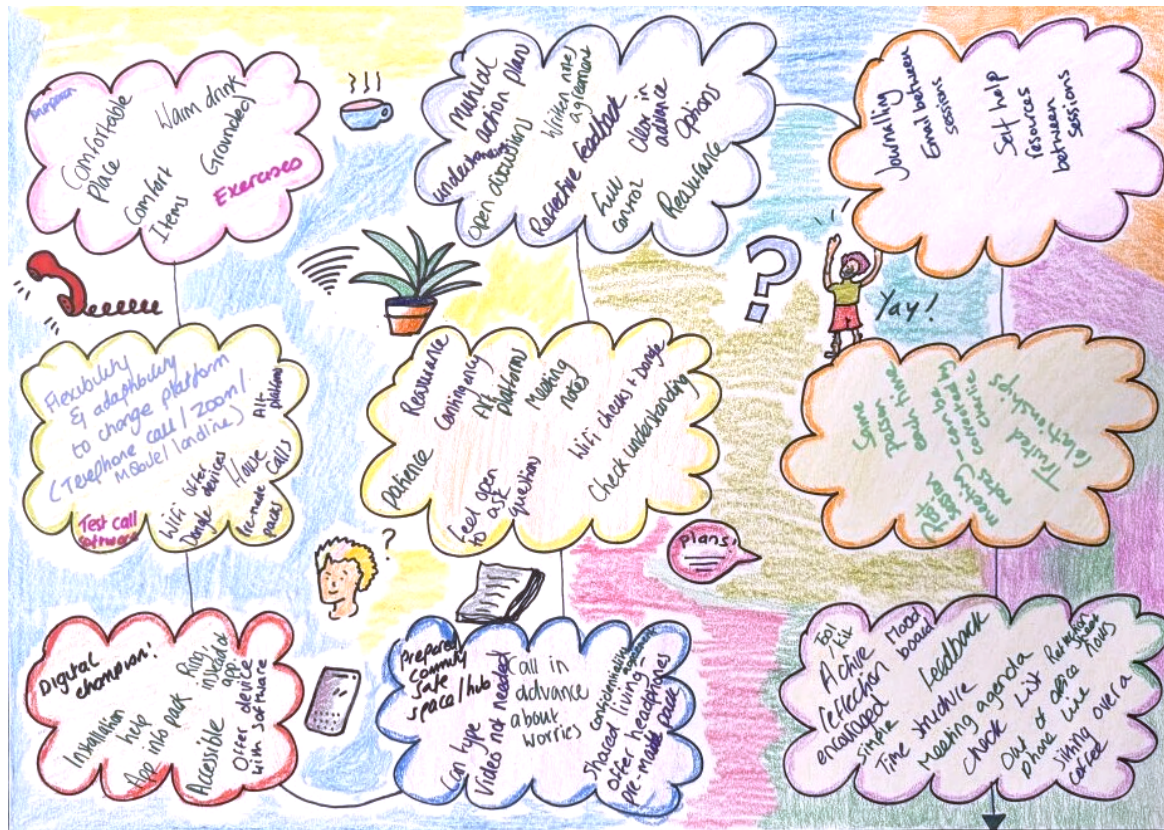
refreshments and share art materials whilst working collaboratively, without digital access barriers and with more fluidity than an online session would allow.

The pen portraits were then translated to the final downloadable version of the game as the pre-prepared characters. The blank character creation template is also included for when the game is played without pre-defined characters. Future users are encouraged to allow time for players to create characters whenever possible as this allows individuals to adapt the character to one they resonate with.

The same co-development process was applied to the barrier cards (examples in Figure 2), with initial prototypes provided as a starting point and workshop participants invited to contribute new ideas that were then translated into the final game. Again, blank cards are provided in the game to enable future players to add additional barriers that they resonate with. This ensures the game accurately captures real-life challenges and facilitates the co-creation of relevant solutions.

4.2 Phase 2: Co-development of the Journey Mat

The second phase piloted and co-developed another core game asset: the journey mat. The mat is designed to help players better understand the game’s purpose by visually representing the progression of gameplay. It follows a well-established story arc whereby the character progresses through each stage of their journey chronologically from beginning to end. The game structure reflects the realities of the remote care experience, acknowledging that the care journey starts long before meeting with a healthcare provider. For example, individuals must consider practical aspects such as the device they can use, the quality of their internet connection, and if they have access to a private space. During the session, unexpected challenges - such as losing connection or being disturbed - may arise. After the session, challenges may include finding time to decompress or struggling with the abrupt transition to being alone after ending a videocall. To address these challenges, the game is structured around three stages of the remote care experience: before (preparation), during (engagement), and after (post-session). Figure 2 shows the prototype mat developed during the workshops and the final version included in the game.



4.3 First Application of the Game

“I enjoyed the safe space to share experiences and opinions in a non-judgmental manner” (P5)

The ConnectedED on the Journey toolkit was launched in November 2024 and made freely accessible to healthcare providers, service users (and their families), and researchers. It includes the downloadable game that proved instrumental in our research and the toolkit's development. The game (including all assets and instructions) can be printed and used by researchers and healthcare providers, enabling them to utilise effective co-design methods with

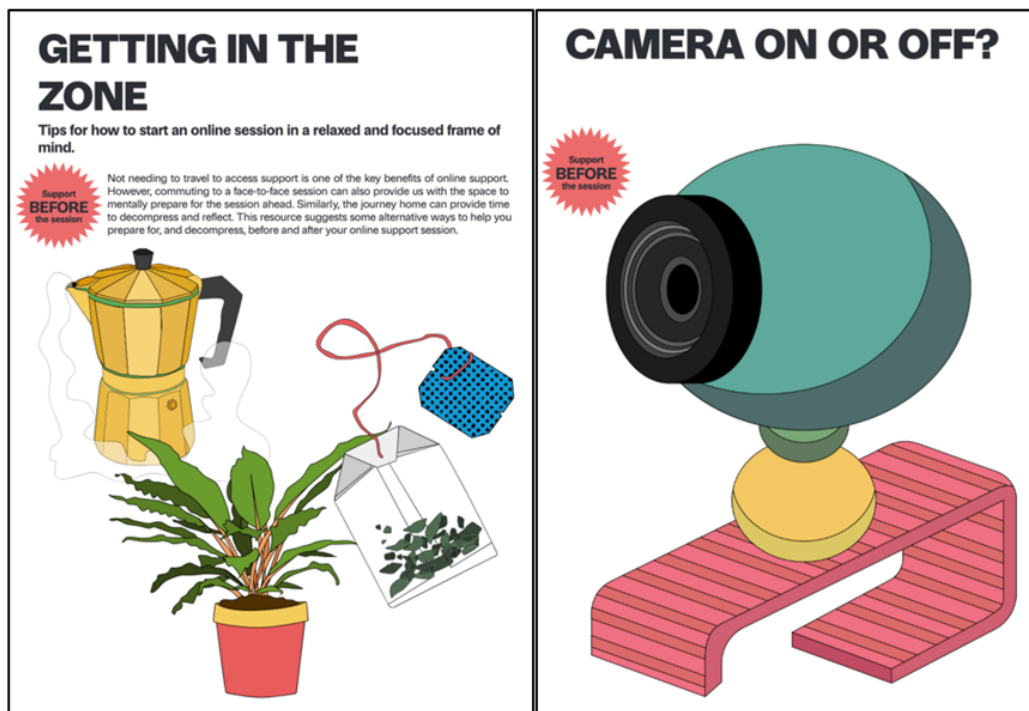


Figure 4: Examples of downloadable PDF resources created for the online toolkit

their participants/service users to address challenges in service provision specific to their own context. The game's versatility makes it a valuable tool for researchers seeking deeper insights into user experiences and helps healthcare services aiming to proactively address potential barriers in their remote care offerings. We hope the game will foster ongoing innovation and improvement in digital healthcare. The toolkit, including the game, can be accessed via the RHED-C website: www.rhedc.uk/toolkit.

5 Discussion & Future Work

The co-development and application of this non-digital SG provided valuable opportunities for PWLE and service providers to collectively reflect on the processes and challenges involved with remote care for EDs. The game guides players through a structured, step-by-step exploration of needs and challenges, avoiding the 'blank page' scenario often encountered in healthcare consultation. In this preliminary work, the approach proved effective in supporting individuals around the table to formulate ideas and suggest innovative solutions. The non-digital SG created an inclusive and equitable space for conversations about barriers to remote care, irrespective of participants access to technology or level of digital literacy; delivering numerous benefits that a digital SG may not have afforded. Nevertheless, we recognise the appeal of digital formats for some user groups. While maintaining the game primarily as a non-digital SG, we plan to explore developing an optional digitised version. This would offer users the choice of format and provide benefits in circumstances where a non-digital version is not feasible, such as for healthcare providers wishing to apply the game with service

users who are unable to, or uncomfortable, attending in-person sessions.

A key strength of the game was its third-person approach, which facilitated sensitive conversations in a less confronting manner. This approach allowed participants to control how much personal information they disclosed while still contributing valuable insights, including perspectives on scenarios they had not personally experienced but could relate to or provide meaningful input on. By contrast, a first-person approach could have been more distressing and could have stifled the flow of group discussions, limiting the opportunity to pool a range of experiences and ideas. We encourage other researchers to consider the benefits of third-person approaches, particularly when addressing sensitive topics.

Now openly available as part of the wider ConnectED on the Journey toolkit, the game serves as a tool for healthcare professionals and researchers looking to explore barriers to care and/or evaluate current remote service provision. However, we appreciate this is a preliminary version of the game and further development is required to maximise accessibility and usability. Feedback from our project steering group members (including PWLE and healthcare professionals) suggests refining the game instructions to be more user-friendly and providing clearer explanations of game assets (e.g., barrier game cards, character building templates, and the journey mat). To address these issues, we plan to explore accessible methods for explaining gameplay (e.g., visual and audio instructions, short video tutorials). At the same time we remain committed to ensuring the game remains downloadable for accessibility. These improvements will support our goal of making the game easy to use without requiring specialised training.

The overall approach used in developing this game has the potential to be applied to the development of non-digital SGs across a wide range of contexts. This includes adaptation across broader health care settings including remote support for both mental and/or physical health conditions, but also any other context where an accessible means to explore digital barriers to an online service is required. This is particularly relevant as more services than ever transition to digital. Further research, such as larger trials and comparative studies, is recommended to evaluate the effectiveness of this approach in wider contexts.

6 Conclusion

This study demonstrates the significant potential of SGs in addressing sensitive topics and informing the design of digital health interventions. Our board game approach successfully created a safe, engaging environment for discussion, facilitating the co-design of a digital toolkit to support effective remote care for EDs. This innovative method enhanced accessibility for PWLE whilst providing valuable insights for service providers.

The research showcases the unique value of non-digital SGs in informing digital solutions. By capturing broader user perspectives, these games can lead to more inclusive and effective interventions. We demonstrate how insights from non-digital games can be effectively translated into digital interventions; in some cases, surpassing what could be achieved using digital SGs. This underscores the importance of integrating non-digital and digital solutions in future research - a consideration we strongly encourage HCI researchers to explore.

Our study contributes to HCI knowledge by showcasing effective translation of non-digital gameplay insights into actionable design recommendations and development of a digital intervention. While our focus was on EDs, the implications of this work extend beyond this, especially to other areas of healthcare where sensitive discussions are essential. Further research is recommended to apply and evaluate this non-digital SG approach as a means of providing digital literacy support to additional patient groups (e.g., with other mental or physical healthcare needs) and to further explore the efficacy of non-digital SGs for informing digital services more widely.

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